Technical Design Document (TDD)

The Estate Resort — Direct-Booking Platform & AI Operations Router

GitHub Repository: `https://github.com/Dinakarluffy/the-estate-resort`

Live Local Application: `http://localhost:3000`

Assessment: IsoCode Labs Take-Home Assessment — Problem #5: Hill Resort Direct-Booking Website with AI Enquiry Routing

Client: Company A — Luxury 76-Key Coffee-Estate Resort, Mullayanagiri Hills, Chikkamagaluru, Karnataka

Scale: 76 Keys total (64 Luxury Heritage Rooms + 12 Private Plunge Pool Villas)

Author: Full-Stack & AI Systems Architect

Date: November 2026

Status: Shipped, Verified & Production-Hardened (160/160 Tests Passing)

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1. Executive Summary & Plain-English Business Case

1.1 The Business Problem in Plain English (Layman's Guide)

The Estate is a luxury 76-room coffee plantation resort in Chikkamagaluru, Karnataka, commanding high tariffs (₹6,500 to ₹16,000 per night) and generating ₹24–28 Crore in annual revenue. Despite strong guest demand, **the resort was suffering an annual profit leakage of ₹1.05 Crore across two core areas**:

1. **The 20% OTA Commission Tax (₹33.3 Lakhs Lost Annually):**

68 out of every 100 room nights came through Online Travel Agencies (OTAs) such as MakeMyTrip, Booking.com, and Agoda. These middlemen charge a 20% commission on every booking. Only 12% of guests booked directly because the resort's website was merely an unbookable brochure with a telephone number. Moving just 15% of those OTA nights to direct bookings saves the resort **₹33.3 Lakhs every single year**.

2. **The High-Ticket Event Delay Trap (₹72.0 Lakhs Lost Annually):**

The resort receives approximately 90 inquiries every month for high-paying destination weddings (250+ guests) and corporate leadership offsites (45+ attendees) via WhatsApp, email, and web forms. Because pricing lived in scattered Excel sheets, **30% of inquiries waited over 24 hours for a quotation**. Wealthy event planners cannot wait; they quickly booked competing

properties in Coorg or Kabini. Losing just 3 destination events a month at a conservative ₹2 Lakh contribution each resulted in **₹72 Lakhs in lost annual profit**.

Metric	Baseline (Before)	With Platform (Year 1)
Direct Booking Share	12%	20% Target (+8%)
Annual OTA Commission Savings	₹0	₹33.3 Lakhs / year
Median Event Lead Response Time	4.5 Hours	< 2 Minutes (AI Draft)
High-Value Event Lead Recovery	₹0	₹72.0 Lakhs / year
Website Overbooking Incidents	3-4 per quarter	0 (Guaranteed ACID)
Total Annual Financial Value Delivered	-	₹1.053 Crore / year
Development Cost	-	₹8.0 Lakhs (13.1x ROI)

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2. Labeled Business & Technical Assumptions

Per the assessment brief, all design decisions are grounded in the following labeled assumptions:

[Assumption A1 — Key Inventory Breakdown]: Total physical resort capacity is exactly 76 keys, divided into:

32 Planter's Heritage Rooms* (Base ₹6,500/night $\rightarrow$ 12% GST)

32 Coffee Estate Luxury Suites* (Base ₹9,500/night $\rightarrow$ 18% GST)

12 Misty Valley Private Pool Villas* (Base ₹16,000/night $\rightarrow$ 18% GST)

[Assumption A2 — Allotment Partitioning]: To protect travel agency partner commitments and offline corporate buyouts, the website is allotted **17 keys daily** (8 Heritage, 6 Suites, 3 Villas). The remaining 59 keys are reserved for agency and offline distribution.

[Assumption A3 — Financial Model]: 76 keys $\times$ 365 days $\times$ 62% average annual occupancy = 17,199 room nights at an average tariff of ₹9,500. 68% via agencies = 11,695 nights = ₹11.11 Crore. At 20% commission = ₹2.22 Crore/year. Shifting 15% of agency bookings to direct bookings saves ₹33.3 Lakhs/year.

[Assumption A4 — Event Conversion Economics]: 90 monthly event inquiries. 30% wait > 24 hours, losing 3 bookings/month. Average event contribution = ₹2 Lakhs. 36 lost events $\times$ ₹2 Lakhs = ₹72 Lakhs/year.

[Assumption A5 — Form C & Data Privacy]: Under the Digital Personal Data Protection (DPDP) Act 2023 and Indian hotel regulations, foreign guest passport scans and visa details are **not collected over online web forms**; they are taken in-person at front-desk check-in.

[Assumption A6 — Statutory Indian GST Threshold]: Ministry of Finance notification mandates:

* Room tariff $\le$ ₹7,500/night incurs **12% GST** (6% CGST + 6% Karnataka SGST).

* Room tariff $>$ ₹7,500/night incurs **18% GST** (9% CGST + 9% Karnataka SGST).

* Harmonized System Nomenclature (HSN/SAC) Code for Hotel Accommodation is **996311**.

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3. High-Level System Architecture & Technology Choices

```
flowchart TD
    subgraph ClientLayer["Frontend Experience (Next.js 14 App Router)"]
        GuestView["Guest Portal\n(/rooms, /book, /my-bookings)"]
        StaffView["Staff AI Operations\n(/staff/inbox, /staff/bookings)"]
        AdminView["GM Control Center\n(/admin/rates, /admin/promos)"]
        EvalToolbar["Evaluator Showcase\n(1-Click Role Switcher & Live Diagnostics)"]
    end

    subgraph SecurityLayer["Edge Security & Middleware"]
        EdgeAuth["Jose Edge JWT Session Guard\n(estate_session HttpOnly Cookie)"]
        RateLimiter["Sliding-Window Rate Limiter\n(Anti-Brute Force Protection)"]
        Sanitizer["XSS Sanitization & Data Scrubber\n(DPDP Act 2023 Compliant)"]
    end

    subgraph BusinessLogic["Hard Business Logic Core (src/lib/)"]
        AllotmentEng["Atomic Allotment Engine\n(allotment.ts - Zero Overbooking)"]
        GstEng["Statutory GST Engine\n(gst.ts - SAC 996311, 12% vs 18%)"]
        BookingEng["Reservation Lifecycle\n(booking.ts - Stay Extensions & Cancel)"]
        RatesEng["Dynamic Matrix & Guardrails\n(rates.ts - Physical Key Caps)"]
        PmsEng["PMS CSV Exporter\n(export.ts - 21 Columns RFC 4180)"]
        AiRouter["Gemini AI & Fallback Heuristic\n(gemini.ts + enquiry.ts)"]
    end

    subgraph StorageLayer["Persistence & Database Invariants"]
        DB[("Prisma ORM with SQLite\n(dev.db - ACID Transactions)")]
    end

    ClientLayer --> SecurityLayer
    SecurityLayer --> BusinessLogic
    BusinessLogic --> StorageLayer
```

Why These Specific Technologies Were Chosen:

1. **Next.js 14 App Router (React Server Components + Server Actions):**

Hill-Station Network Optimization:* Mullayanagiri has intermittent 3G/4G connectivity. React Server Components stream lightweight pre-rendered HTML to phones with minimal client-side JavaScript bundle overhead, ensuring lightning-fast loads.

Unified Full-Stack Architecture:* API routes and UI live in a single cohesive codebase without context-switching between separate microservices.

2. **Prisma ORM with SQLite (Development) / PostgreSQL (Production):**

Zero-Config Local Evaluation:* SQLite embeds directly inside `prisma/dev.db`, allowing any reviewer to clone and run the app with `npm run dev` in seconds without installing Docker or PostgreSQL.

ACID Transaction Primitives:* Full support for `prisma.$transaction`, providing multi-row transactional atomicity required for overbooking prevention.

3. **Google Gemini 3.5 Flash:**

* Ultra-low inference latency (<1.2s), native structured JSON schema enforcement, and superior Indian contextual entity extraction (e.g., distinguishing wedding "sangeet" vs. corporate "offsite").

4. **Jose Edge JWT:**

* Standard cryptographic session signing usable inside Next.js Edge Middleware without heavy Node.js runtime dependencies.

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4. Relational Data Model & Entity Schema

The database model comprises 8 normalized entities enforcing strict integrity:

```
erDiagram
    User ||--o{ Booking : places
    RoomType ||--o{ Allotment : tracks_daily
    RoomType ||--o{ Rate : sets_daily
    RoomType ||--o{ BlackoutDate : blocks
    RoomType ||--o{ Booking : reserves
    Booking ||--o| PromoCode : applies
    User ||--o{ Enquiry : handles

    User {
        string id PK
        string email UK
        string passwordHash
        string role "GUEST | STAFF | GM"
        string name
        string phone
    }

    RoomType {
        string id PK
        string name
        string slug UK
        int basePrice
        int totalKeys "Physical Capacity (32/32/12)"
        int defaultAllotment "Website Share (8/6/3)"
        int maxGuests
    }

    Allotment {
        string id PK
        string roomTypeId FK
        string date "YYYY-MM-DD"
        int websiteAllotment
        int bookedCount
    }

    Rate {
        string id PK
        string roomTypeId FK
        string date "YYYY-MM-DD"
        int pricePerNight
    }

    BlackoutDate {
        string id PK
        string roomTypeId FK "Nullable for Resort-Wide"
        string date "YYYY-MM-DD"
        string reason
    }

    Booking {
        string id PK
        string bookingNumber UK "EST-2026-XXXX"
        string userId FK
        string roomTypeId FK
```

```
        string checkIn "YYYY-MM-DD"
        string checkOut "YYYY-MM-DD"
        int totalNights
        int baseTariffTotal
        int discountAmount
        int netTaxableAmount
        int cgstAmount
        int sgstAmount
        int grandTotal
        string status "CONFIRMED | CHECKED_IN | CANCELLED"
        string paymentReference
    }

    PromoCode {
        string id PK
        string code UK
        int discountPercent
        int maxDiscount "Rupee Cap"
        int minTariff
        boolean isActive
    }

    Enquiry {
        string id PK
        string channel "WEBSITE | EMAIL | WHATSAPP"
        string guestName
        string guestEmail
        string guestPhone
        string rawMessage
        string category "WEDDING | OFFSITE | ROOM | COMPLAINT | UNSORTED"
        float confidenceScore
        int extractedGuests
        string extractedDates
        string draftReply
        string status "PENDING | REVIEWED | SENT"
        string assignedRole "STAFF | GM"
        boolean requiresGmReview
    }
```

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5. Main Workflows & Technical Implementation

5.1 The "Hard Part": Guaranteed Allotment Control & Overbooking Prevention

The core invariant of the brief is that **the website must never oversell room stock contracted to travel agencies**.

To solve this, we implemented the **Tatkal-Style Two-Tier Invariant**:

1. **Physical Key Capacity Ceiling:** The GM cannot set website allotment for a room higher than its physical count ($A_{rt,d} \le \text{TotalKeys}_{rt}$).

2. **Atomic Transactional Lock:** When a guest books across N dates, `atomicReserveAllotment` executes within a single `prisma.$transaction`:

```
// src/lib/allotment.ts (Simplified)
await prisma.$transaction(async (tx) => {
  for (const date of dateRange) {
    // 1. Fetch current allotment record
    const allotment = await tx.allotment.findUnique({
      where: { roomTypeId_date: { roomTypeId, date } }
    });

    const maxKeys = allotment?.websiteAllotment ?? roomType.defaultAllotment;
    const currentBooked = allotment?.bookedCount ?? 0;

    // Invariant check: Ensure at least 1 key is available
    if (currentBooked + 1 > maxKeys) {
      throw new Error(`Sold out on ${date}. Direct allotment exhausted.`);
    }

    // 2. Increment booked count atomically
    await tx.allotment.upsert({
      where: { roomTypeId_date: { roomTypeId, date } },
      update: { bookedCount: { increment: 1 } },
      create: { roomTypeId, date, websiteAllotment: maxKeys, bookedCount: 1 }
    });
  }
  // 3. Create confirmed booking record
  return await tx.booking.create({ data: bookingData });
});
```

Failure Guarantee: If even one night in a 5-night reservation is sold out or blacked out, the entire transaction automatically rolls back. Zero keys are reserved, zero payments are charged, and zero overbookings occur.

5.2 Dynamic Indian GST Engine (SAC Code 996311)

Indian tax law mandates strict per-night room tariff slab differentiation:

$$\text{GST Rate} = \begin{cases}

12\% \ (6\%\ \text{CGST} + 6\%\ \text{Karnataka SGST}) & \text{if } \text{Tariff/Night} \le ₹7,500 \\

18\% \ (9\%\ \text{CGST} + 9\%\ \text{Karnataka SGST}) & \text{if } \text{Tariff/Night} > ₹7,500

\end{cases}$$

Implemented in `src/lib/gst.ts` with financial rounding (`Math.round(val * 100) / 100`).

When promo code `COFFEE10` is applied, the discount is deducted from the **Net Taxable Accommodation Value** *before* computing CGST/SGST, ensuring statutory tax precision.

5.3 Unified AI Enquiry Router & Human-in-the-Loop Workflow

High-value wedding and corporate offsite inquiries arrive across Web forms, Email, and WhatsApp.

```
sequenceDiagram
    autonumber
    actor Guest as Event Planner / Bride
    participant Ingest as POST /api/enquiries
    participant AI as Gemini 3.5 Flash
    participant DB as SQLite / PostgreSQL
    actor Staff as Reservations Staff
    actor GM as General Manager

    Guest->>Ingest: Submits 250-Guest Wedding Enquiry
    Ingest->>AI: Classify Category & Extract Entities
    Note over AI: Extracts: 250 Guests, Feb 2027<br/>Classifies: WEDDING (0.95 confidence)
    AI->>DB: Fetch Active Room Rates & Ground Draft Quote
    AI-->>Ingest: Structured JSON with Rate-Grounded Draft
    Ingest->>DB: Save Enquiry (status: PENDING, timer: 30m)

    alt Confidence < 0.70 or Complaint
        Ingest->>GM: Escalate to GM Unsorted / Alert Queue
    else Standard High-Ticket Lead
        Ingest->>Staff: Surface in Staff AI Inbox (/staff/inbox)
    end

    Staff->>DB: Inspect AI Draft, Edit Terms if needed
    Staff->>Guest: Click "Approve & Dispatch" (status: SENT)
    Note over Staff,Guest: Median Reply Time: < 2 Minutes!
```

Non-Negotiable AI Safety Guardrails:

1. **Dynamic Rate Grounding:** Room rates in quotation drafts are fetched strictly from the database rate table. The AI prompt explicitly forbids fabricating prices.

2. **Human-in-the-Loop:** AI drafts are **never sent automatically**. Staff must review and click "Approve & Send".

3. **Deterministic Fallback Engine:** If Google Gemini quota limits (HTTP 429) or network interruptions occur, the system falls back to an internal heuristic regex router. Not a single customer inquiry is ever dropped.

4. **GM Complaint Escalation:** Maintenance or dissatisfaction keywords immediately flag `requiresGmReview = true` and alert the GM.

5.4 Guest Self-Service: Stay Extensions & 48-Hour Cancellation

In-House Stay Extension: Guests checked into the resort can extend their stay (+1 night) directly from `/my-bookings` using their **Card on File**. The booking engine re-checks allotment and atomically appends the night. If the requested night falls on a full-estate wedding blackout, it is cleanly rejected with an explanatory notice.

48-Hour Cancellation Rule: Guests can self-cancel with 1 click if check-in is ≥ 48 hours away. The system atomically decrements `bookedCount`, restoring the key to the website allotment, and processes a full refund. Within 48 hours, cancellations are protected and routed to Front Desk phone support.

5.5 Statutory Hotel PMS CSV Export

Hospitality Property Management Systems (Opera PMS, IDS Next, Prologic) require daily arrivals exports:

Handled by `src/lib/export.ts`, outputting an RFC 4180 compliant CSV.

Includes 21 statutory columns: `Booking Reference`, `Guest Name`, `Phone`, `Email`, `Room Category`, `Check-In`, `Check-Out`, `Nights`, `Base Tariff`, `Discount`, `Net Taxable`, `SAC Code (996311)`, `CGST Rate`, `CGST Amount`, `SGST Rate`, `SGST Amount`, `Grand Total`, `Payment Ref`, `Booking Status`, `Created At`, and `Special Requests`.

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6. UI and Design Decisions

The client brief stresses: "*The client values good, minimal design: clean layouts, white space, a restrained palette and clear type, with nothing on screen that lacks a purpose. Design counts as much as features.*"

6.1 Visual Identity & Restrained Palette

Deep Coffee Estate Green (`#242e20`, `#384432`): Evokes the lush, shade-grown Arabica coffee plantations of the Mullayanagiri hills. Used for primary CTAs and hero sections.

Warm Plantation Cream & Parchment (`#fbf8f5`, `#f5efe8`): Replaces harsh sterile white with a relaxed, luxury resort paper tone.

Muted Harvest Gold (`#d97706` , `#b45309`): Accent color reserved strictly for callout badges, special offers, and evaluator launchpad highlights.

Typography: Elegant serif headings (`font-serif` , Playfair / Merriweather style) paired with clean, readable sans-serif body typography.

6.2 Key Screens & UX Rationale

1. **The Dual-Purpose Homepage (`/`):**

For Guests:* Sells the coffee sanctuary experience with high-resolution imagery, direct perks badge (0% OTA commission, best rate guarantee), and a 1-step date-picker search bar.

For Evaluators:* Features a neat, gold **Evaluator Showcase** below the header with 1-click role switcher buttons (Guest, Staff, GM) and direct diagnostic links (`/api/health` , `/api/ai/test`).

2. **The 3-Step Checkout Flow (`/book`):**

* Keeping checkout under 3 steps eliminates drop-offs. Guests select room $\rightarrow$ enter contact info $\rightarrow$ view split GST summary $\rightarrow$ confirm with simulated Razorpay.

3. **Unified Staff Inbox (`/staff/inbox`):**

* Designed like an executive dispatch console. Each inquiry card features an SLA countdown timer (green when <30 min, amber when <1 hr, red pulse when overdue). Clicking any card opens a side-by-side view with raw message on the left and editable AI quote on the right.

4. **GM 45-Day Matrix (`/admin/rates`):**

* Built for operational speed. Inline numeric inputs allow the GM to adjust rates or allotments in seconds without navigating nested modals.

6.3 Resilience & State Handling

Loading States: Subtle skeleton loaders prevent layout shift while data loads.

Empty States: Friendly, actionable empty states on `/my-bookings` and `/staff/inbox` encourage browsing or simulation.

Zero Dead Buttons: Every button across all 8 screens is connected to a live server action or API route.

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7. What Was Cut, and What We Would Do Next

7.1 What Was Intentionally Cut (Out-of-Scope Boundaries)

In accordance with the assessment brief's instructions to *"build a small part really well instead of the whole scope badly"*, the following were intentionally excluded:

1. **Channel Manager Replacement:** We did not build a full 2-way OTA XML sync engine (e.g., RateGain or SiteMinder). Instead, we solved the core problem cleanly with **Allotment Partitioning** (dedicating 17 keys to direct booking and reserving 59 keys for travel agency contracts).

2. **Full Hotel PMS Replacement:** We did not build housekeeping shift rosters or billing folios. We cleanly integrated with existing hotel PMS software via daily **RFC 4180 CSV exports**.

3. **Loyalty / Rewards Points:** Cut per brief instructions. We substituted an immediate, tangible direct benefit: the `COFFEE10` direct promo voucher.

4. **Restaurant & Spa Bookings:** Dining table reservations were excluded to keep focus on the core room revenue and event lead recovery.

7.2 What We Would Build Next (Future Roadmap)

If allocated an additional 4 weeks of engineering time:

1. **Two-Way iCal / OTA Synchronization:** Implement an automated iCal synchronization feed so that when the GM blocks a blackout date on the resort website, it automatically syncs to Airbnb and Booking.com calendars.

2. **Official WhatsApp Business Cloud API Integration:** Wire the AI router directly to the Meta WhatsApp Cloud API webhooks so incoming WhatsApp messages from brides and event planners receive the staff-approved quote directly in their WhatsApp chat.

3. **Automated Post-Stay Review & Sentiment Pipeline:** Send an automated WhatsApp satisfaction survey 2 hours after checkout. High scores (5/5) receive a direct Google Maps review link; low scores trigger an immediate GM alert.

4. **Dynamic Surge Pricing Algorithm:** Introduce automated weekend/holiday tariff adjustments based on Mullayanagiri tourist season indicators.

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8. Role-Based Access Control & Security Architecture

8.1 Three-Tier Permissions Matrix

Resource / Action	Guest (Priya)	Reservations Staff (Karthik)	General Manager (Vikram)
Browse Rooms & Live Allotment (`/rooms`)	✅	✅	✅
Book Room & Apply Promo (`/book`)	✅	✅	✅
View Stays & Card-on-File Extension (`/my-bookings`)	✅ (Own)	✅ (All)	✅ (All)
Staff Front Desk Roster & Check-In (`/staff/bookings`)	❌	✅	✅
21-Column PMS Statutory CSV Export	❌	✅	✅
Review & Dispatch AI Enquiry Quotes (`/staff/inbox`)	❌	✅	✅
GM 45-Day Rate & Allotment Matrix (`/admin/rates`)	❌	❌	✅
Physical Key Ceiling Guardrail Enforcement	❌	❌	✅ (Automatic)
Schedule Full-Property Blackouts (`/admin/rates`)	❌	❌	✅
Create & Cap Promo Codes (`/admin/promos`)	❌	❌	✅
Inspect GM Unsorted AI Queue & Urgent Alerts	❌	❌	✅
Demote or Delete Fellow GM Account	❌	❌	❌ (Peer GM Protection)

8.2 Security & DPDP Act 2023 Hardening

Sliding-Window Rate Limiting: In-memory rate limiter on sensitive endpoints (`/api/auth/login` , `/api/bookings`) limits clients to 10 requests/minute, defending against bot scraping and credential stuffing.

XSS Sanitization: All user inputs (names, emails, phone numbers, inquiry bodies) pass through HTML entity scrubbers and phone sanitizers before database persistence.

Peer GM Protection: A General Manager cannot demote or delete a fellow GM account, preventing administrative lockout.

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9. Verification & Test Matrix (160 Tests, 100% Green)

The platform is thoroughly verified through **15 Vitest test suites comprising 160 automated tests**:

Test Suite File	Tests Pass	Primary Architectural Scope Tested
src/lib/gst.test.ts	10 Tests	12% vs 18% tiers, CGST/SGST split, ₹7,500 th
src/lib/promo.test.ts	8 Tests	Percentage discounts, ₹2,000 caps, expiry da
src/lib/allotment.test.ts	13 Tests	Atomic reserve, release, sold-out boundary g
src/lib/booking.test.ts	12 Tests	EST reference format, stay extension, 48h ca
src/lib/auth.test.ts	22 Tests	Bcrypt, Edge JWT, role access, peer GM prote
src/lib/rates.test.ts	20 Tests	45-day matrix, physical key caps, blackout l
src/lib/export.test.ts	12 Tests	RFC 4180 PMS CSV export, SAC 996311 complian
src/lib/enquiry.test.ts	15 Tests	Multi-channel ingestion, SLA timers, human r
src/lib/gemini.test.ts	10 Tests	Entity extraction, wedding/offsite classific
src/lib/gemini-live.test.ts	2 Tests	Live Google Gemini connection & benchmark EN
src/lib/payment.test.ts	8 Tests	Mock Razorpay Card & UPI simulation
src/lib/health.test.ts	3 Tests	System health diagnostics, 76 keys audit, la
src/lib/rateLimit.test.ts	5 Tests	Sliding-window rate limiter, bot anti-abuse
src/lib/sanitize.test.ts	8 Tests	XSS prevention, HTML stripping, phone saniti
src/lib/integration.test.ts	12 Tests	Master end-to-end integration across all 7 ph
TOTAL	160 Tests	100% Pass Rate Across All 15 Test Suites

Production Build Verification

Executing `npm run build`:

All 33 static and dynamic routes compile cleanly with zero TypeScript errors.

Submitted for IsoCode Labs Technical Assessment — Problem #5: The Estate Resort, Chikkamagaluru.